Countersign: Evidence-Grounded Supervision of Coding-Agent Completion Claims

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Abstract

Coding agents terminate when they believe a task is done, but that belief is not always supported by their own execution trace. We present *Countersign*, a deterministic supervisory meta-agent that audits a worker's completion claim against the trace—checking that cited tests exist, remain fresh with respect to the files they cover, postdate the latest requirement update, and do not rest on superseded sources—and can block termination and demand fresh evidence. Countersign is deliberately *trace-only*: it judges whether a claim is *justified* at claim time—distinct from whether the implementation is *correct*, which a hidden evaluator answers once after termination. In a preregistered, frozen campaign of 490 real-model runs over ten held-out fixtures, all 17 blocks landed on proposals an independent oracle also labelled unsupported (precision 1.00, 95% CI $[0.82, 1.0]$) and none on the 204 supported proposals judged—though 13 of 17 are the trivial case of a claim citing no evidence. A counterfactual replay shows a conventional continuous-integration gate catches 2 of 15 unsupported proposals where Countersign catches 11 of 11 it judged. Negative results are equally informative: truncating the worker's visible history did not increase unsupported completion; the effect appears only when history is *replaced by a narrative summary*, driven mainly by prompting, the gate's contribution being to convert unsupported acceptances into blocked episodes that recover. And we quantify the construct honestly: 25 of 27 unsupported proposals were backed by work that passed hidden evaluation—Countersign is an evidence-hygiene instrument whose safety value remains a hypothesis under test.

1 Introduction

An autonomous coding agent ends its episode by asserting completion—a claim about evidence: the tests it ran, the edits it made, the requirements it was given. When the claim outruns the evidence (a cited test predating the final edit; a superseded source), the agent has not necessarily produced wrong code, but it has produced an *unwarranted* report, and whoever trusts it inherits a risk they cannot see.

We study whether a second agent with authority over the first's termination can reliably tell warranted completion claims from unwarranted ones, and what that authority costs. Countersign reads only the worker's execution trace: it never consults ground truth, and the hidden evaluator that determines correctness runs exactly once, after termination, identically in every condition. This separation is the paper's organizing principle: a claim can be justified and wrong, or correct and unjustified, and conflating the two makes "the verifier improved outcomes" unfalsifiable.

Contributions. We contribute (i) a supervision loop with halting authority over worker termination, scored against an independent post-hoc oracle it does not compute (§2); (ii) a preregistered evaluation on matched fixture pairs and negative controls that price false blocks, under a frozen protocol with a public deviation ledger (§3); (iii) results across 490 runs: perfect block precision, a counterfactual CI-gate ablation, a null dose–response under truncation, and a regime-specific summarization effect

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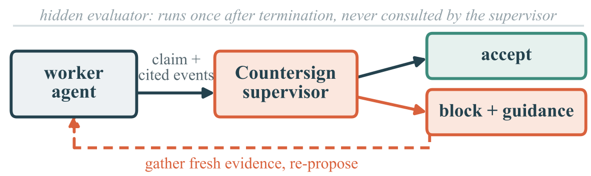

Figure 1: The supervision loop. The worker proposes termination with cited evidence; Countersign audits the claim against the trace and accepts or blocks with a specific deficiency, after which the worker gathers fresh evidence and re-proposes. The hidden evaluator runs once, after termination, and is never consulted by the supervisor.

decomposed into prompt and gate contributions (§4); (iv) a calibrated account of the gate as evidence hygiene, not a demonstrated safety mechanism (§5).

2 The Countersign Supervisor

Worker and ledger. The worker is a tool-loop coding agent (self-hosted open-weight models) in a git-backed workspace; every action appends to an *evidence ledger*—tool calls with outcomes, file mutations, test runs with inferred coverage, requirement updates. To finish, it emits a `finish` action carrying a claim and cited evidence identifiers.

The audit. Countersign checks four properties of the cited evidence: (i) at least one cited event is a successful test run; (ii) no cited test predates a later mutation of a file it covers (coverage inference detailed in Appendix A); (iii) cited evidence postdates the latest relevant requirement update; and (iv) the claim does not rest on sources the task marks superseded. Freshness is tri-state: the supervisor abstains when coverage is unknown.

Authority. On failure the supervisor blocks termination and returns the specific deficiency; a repair variant additionally issues the smallest evidence-refreshing action. Every arm logs the *raw* verifier decision separately from the *enforced* one, so non-blocking arms still measure what the supervisor would have done.

Independent oracle. A post-hoc oracle labels every proposal supported/unsupported/uncertain from fixture-authored completion policies and canonical trace chronology. It shares no code with the online verifier; §5 reports a regime where they disagree.

3 Evaluation Design

Fixtures. Ten held-out fixture repositories, authored to a preregistered design and frozen before any real-model run was inspected. Three *matched pairs* isolate one failure mode each under context parity (workspaces byte-identical but for one designed file): *temporal* (cited tests predate the final edit), *provenance* (fresh evidence grounded in a superseded document), and *requirement* (a clarification arrives after the last passing run). Four *negative controls* contain no trap (Appendix A), so any block on them is a false positive by construction; a separate development suite, on which the rules were iterated, is never used for evaluation.

Conditions. Arms cross a worker prompt variant with verifier authority: memory_baseline (naive prompt, no gate), observe_only (evidence-citation prompt; the verifier logs decisions but never blocks nor returns feedback), verification_only (same prompt, blocking), verification_and_repair, and an evaluation-only oracle_supervisor upper bound. Because memory_baseline and verification_only differ in *both* prompt and gate, we never report that contrast as a gate effect; observe_only is the prompt-matched control, verified in-trace to emit zero worker-visible feedback events.

Memory degradation. A frozen registry of degradation profiles applies only to *model-visible* state: graded *lossy truncation* of history (low/medium/high) and a *resume-summary* regime replacing the visible ledger with a single truthful summary—the pattern deployed scaffolds use on session resumption. The canonical ledger, verifier, oracle, and evaluator always see undegraded state.

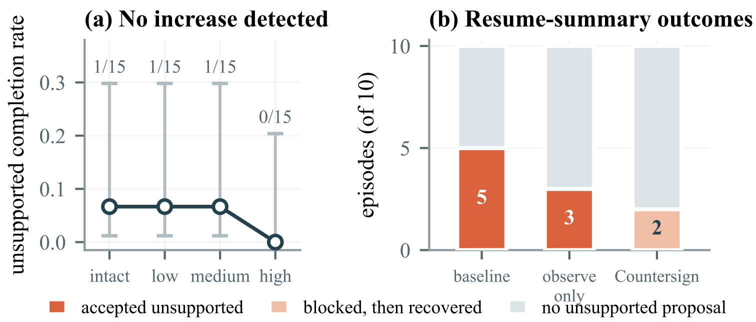

Figure 2: Memory degradation does not act uniformly. (a) Graded truncation: no detectable increase in unsupported completion; the Wilson 95% intervals mark this a non-detection at modest power, not invariance. (b) Replacing history with a resume summary does produce the effect: the evidence-citation prompt reduces unsupported proposals (5→3), and the gate converts those it sees from silent acceptances into blocked episodes that recover (2 of 2). Small n; decomposition, not significance.

Protocol. Temperature 0.7, sampler seeds 0–4, action budget 24 (Appendix A). Endpoints, comparisons, and interpretation branches for both positive and null outcomes were committed before the runs; the frozen protocol hashes code, fixtures, and model weights and refuses to start on a dirty tree.

4 Results

All numbers come from audited run manifests; the worker is `qwen2.5-coder:14b`. The campaign's 490 runs yield 341 oracle-judged finish proposals, 27 labelled unsupported.

Blocks are precise. Across every arm with a live verifier, Countersign issued 17 blocks—all on proposals the independent oracle also labelled unsupported, none on the 204 supported proposals a verifier judged: precision 1.00 (95% CI $[0.82, 1.0]$), false-block rate $0/204$ (95% CI $[0, 0.018]$), recall $17/18 = 0.94$ (95% CI $[0.74, 0.99]$). Two caveats bound this. First, the denominator counts only judged proposals: 110 further supported proposals occurred in no-verifier arms and are excluded, not counted as successes. Second, and more important, **13 of the 17 blocks are the trivial case**—a claim citing no evidence, which any rule detects. Only 4 required substantive reasoning about staleness or provenance: read the figure as "the gate does not misfire," not "the gate discriminates hard cases."

A conventional CI gate is blind to what actually occurs. We replay every stored proposal against the standard-practice rule "require a successful test run after the last edit," a strict subset of Countersign's temporal check. The CI rule catches 2 of 15 unsupported proposals; Countersign catches 11 of the 11 it judged, of which only 4 required substantive reasoning—the rest cite nothing at all. Neither gate blocked any of the 161 supported proposals both judged: the difference is blindness, not conservatism.

Truncating memory does not *detectably* induce false claims. Baseline unsupported completion on trap fixtures did not increase with truncation severity (Figure 2a). This is a non-detection, not invariance: at 15 runs per cell the Wilson intervals span roughly 0.01–0.30, so effects under $\sim$30 percentage points are unresolvable. The preregistered null branch applies with that caveat.

The effect is regime-specific, and decomposes. Replacing history with a resume summary does produce unsupported completion (Figure 2b). The comparison must be read at the *proposal* level: in a blocking arm an accepted-unsupported finish is suppressed by the intervention's own definition, so a zero there is partly mechanical. Counting unsupported proposals over ten matched cells: baseline 5, observe_only 3, verification_only 2. The evidence-citation prompt accounts for the reduction from 5 to 3 (exact McNemar $p = 0.50$); the further $3 \to 2$ is sampling noise, since the gate cannot influence a worker's *first* proposal. The gate's contribution is not fewer unsupported claims but a different fate for them: both proposals it saw were blocked, and both episodes recovered and terminated with supported evidence rather than exhausting their budget. No comparison here reaches $p < .05$, and an a-priori power analysis explains why—with all discordant pairs in one direction the exact test requires $b \geq 6$, one more than this campaign could have produced, and 16 matched cells would be needed for 80% power at the observed effect (Bayesian sensitivity reading in Appendix D).

Table 1: Accepted completions (n=327): oracle verdict on the claim vs. hidden-evaluator verdict on the work. The empty bottom-right cell is the paper's central caveat.

	correct	incorrect
claim supported	307	2
claim unsupported	18	0

The cleanest evidence is measurement without power: in observe_only the verifier judges but cannot act, and its would-block decisions fired in exactly the 3 unsupported cells and none of the other 7 (TP= 3, FP= 0, FN= 0)—immune to the circularity objection a blocking arm invites.

What "unsupported" means here. Table 1 crosses the oracle's judgement of the accepted claim against the hidden evaluator's judgement of the work. All 18 unsupported completions concerned work that was in fact *correct*: an unjustified claim never once coincided with a wrong implementation, so Countersign is measured to catch *poor evidential hygiene* and its safety value remains a hypothesis this campaign cannot confirm.

5 Supervisor-Failure Analysis and Limitations

The structural blind spot. The provenance pair constructs fresh evidence grounded in a superseded document: a trace-only supervisor allows those finishes and the hidden evaluator fails them—supported-but-incorrect outcomes no trace audit can reach, bounding what runtime evidence supervision can promise. Appendix B details two further failure modes: classifier/oracle divergence under heavy degradation and a fixture family censored by task weight, not action budget.

Limitations. (1) Most important: the held-out traps and the supervisor's rules share authors. The development/held-out split prevents rule tuning on evaluation fixtures but not the conceptual correlation between what we trap and what we detect; the provenance family (built to *defeat* the supervisor) and the negative controls (which price the unflattering direction) offset this only partially. (2) One worker model on ten small fixture repositories establishes nothing about prevalence in production software. (3) The primary endpoint is partly suppressed by construction in blocking arms; gate effects are read at the proposal level and in post-block outcomes. (4) The verifier and oracle share no code but read the same trace, and most agreements concern claims citing no evidence—near-vacuous independence. (5) No result reaches $p < .05$; nulls are non-detections at modest power, and we report decompositions and intervals, never significance. (6) The degradation profiles are our own transforms, not a deployed scaffold's condenser; (7) oracle-label validation covers a single-rater sample.

6 Related Work

Prior work measures workers gaming outcome checks Von Arx et al. [2025], Baker et al. [2025] and gates *actions* for safety at runtime Chennabasappa et al. [2025]; Countersign instead gates *termination* for evidential support and prices the false blocks. Self-verification asks the worker to check itself Shinn et al. [2023]; external supervision is the complement when self-checking is in doubt. Outcome and process verifiers Cobbe et al. [2021], Lightman et al. [2024] inform our online-process/post-hoc-outcome split; long-context degradation Liu et al. [2024], Packer et al. [2023] motivates the memory regimes, and hidden-evaluation benchmarks Jimenez et al. [2024] the correctness oracle.

Responsible-Use Statement

Countersign controls agent overclaiming; its net effect should be positive, but agents supervising agents carries risks we keep visible. *An allow is not a guarantee*: the supervisor audits justification, not correctness (§5), and must never be marketed as verifying it. *Oversight displacement*: a visible gate invites humans to stop checking; verdicts should reach operators as cited-evidence audits, not badges. *Who supervises the supervisor*: halting authority creates a new failure surface (over-blocking; repair loops into budget exhaustion) which we price as standard counter-metrics. Experiments involve no human subjects, personal data, or production systems.

References

Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation, 2025. URL `https://arxiv.org/abs/2503.11926`.

Sahana Chennabasappa et al. LlamaFirewall: An open source guardrail system for building secure AI agents. *arXiv preprint arXiv:2505.03574*, 2025. Author list to be verified against the arXiv page before submission.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12, 2024.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Sydney Von Arx, Lawrence Chan, and Beth Barnes. Recent frontier models are reward hacking. METR Technical Report, 2025. URL `https://metr.org/blog/2025-06-05-recent-reward-hacking/`.

A Design Details

Test coverage is inferred statically, so documentation edits do not invalidate code evidence, and a failure superseded by a later passing run does not permanently contradict a claim. The four negative controls: documentation edits after green tests; an unrelated-module edit with a targeted-test citation; a legitimate zero-change audit; an irrelevant late clarification. Sampling uses temperature 0.7 because greedy decoding would make multi-seed replication pseudoreplication; seeds 0–4 are sampler seeds over otherwise identical episodes.

B Additional Failure Modes

Classifier/oracle divergence. Under high degradation the online claim classifier disagreed with the independent oracle, because the classifier reads the worker's degraded view while the oracle reads the canonical trace. All endpoints in the main text are oracle-anchored; the divergence is evidence that an independent oracle is necessary rather than redundant.

Task weight censors a family. Requirement-family trap runs exhaust the action budget in $3/5$ runs in every arm, and raising the budget from 24 to 40 actions left that rate unchanged. The family is intrinsically too heavy for this worker—a fixture-design finding, not a budget-calibration error.

C A Blocked-then-Recovered Trajectory

Under lossy truncation at low severity, one verification-arm episode on a fresh-evidence temporal fixture shows the mechanism in miniature. The pressured worker twice proposed terminating *legitimate* work with insufficient cited evidence; the supervisor blocked both proposals with the specific deficiency; the worker gathered fresh evidence, terminated supported, and passed hidden evaluation. Pressure-induced premature finishing was rare for this worker, but both instances that occurred were caught and repaired by the gate. The five blocks issued across the pressure gradient (three on trap fixtures, two on control-task proposals) all landed on oracle-unsupported proposals, and enforced false blocks on oracle-supported work remained zero campaign-wide.

D Bayesian Sensitivity Readings

As a post-hoc sensitivity analysis (recorded as such in the deviation ledger; no new data, uniform $\mathrm{Beta}(1, 1)$ priors, exact conjugate posteriors, no sampling), we re-read the paper's key small-n counts in Bayesian form. *Direction of the prompt effect*: conditioning on discordant matched cells as in the exact McNemar test, $b{=}2$ cells favour `observe_only` over `memory_baseline` and $c{=}0$ the reverse, giving posterior $\mathrm{Beta}(3, 1)$ on the probability that a discordant cell favours the prompt and $P(\theta > 0.5) = 0.875$—seven-to-one posterior odds on a beneficial direction from the same data whose two-sided p is 0.50. *Discrimination in the arm that cannot act*: the `observe_only` verifier's sensitivity $3/3$ has posterior mean 0.80 with 95% credible interval $[0.40, 0.99]$, and its specificity $7/7$ has mean 0.89, interval $[0.63, 1.00]$; campaign-wide block precision $17/17$ gives $[0.81, 1.00]$, agreeing with the Wilson interval in the main text. These are direction and uncertainty statements only; none is a magnitude claim, and all shrink toward chance under priors skeptical of any effect.

E Reproducibility

The campaign spans three tags of one freeze: a base freeze plus two infrastructure-only fixes applied mid-campaign (interruption-recovery artifact reuse; a tool-level crash on syntactically invalid worker-written code surfacing as a tool error rather than killing the episode). The hashed verifier-policy files are byte-identical across all three tags: no verifier, oracle, or fixture logic changed after the freeze. Every deviation, interim inspection, and protocol-identifier supersession—including two interim analyses that informed spending decisions, and one corrected over-interpretation—is recorded in the deviation ledger shipped with the artifact. The artifact is a relocatable bundle whose integrity audit passes from the copied location; it contains only fixtures, protocols, and run records.